MIL-T-13321B(MU)

10 SEPTEMBER 1963

SUPERSEDING
MIL-T-13321A(ORD)
16 DECEMBER 1959

MILITARY SPECIFICATION

TELESCOPES, ELBOW: M62 SERIES

1. SCOPE

1.1 This specification covers a series of small elbow telescopes used for direct and indirect fire as a component with various sightunits. The individual telescope has a designation affixed to the model number to identify the reticle pattern employed and minor design variations.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-O-13830 — Optical Components for Fire Control Instruments; General Specification Governing the Manufacture Assembly and Inspection of

MIL-F-13926 — Fire Control Materiel; General Specification Governing the Manufacture and Inspection of

MIL-P-14232 — Parts, Equipment and Tools for Ordnance Materiel: Packaging of

MIL-I-45208 — Inspection Requirements, General Specification for

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109 — Quality Assurance Terms and Definitions

DRAWINGS

U. S. ARMY MUNITION COMMAND

A7579975 — Telescope, Elbow: M62

F7649234 — Telescope, Elbow: M62E2 and M62-A1C

FSC 1240

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F8572120 — Telescope, Elbow: M62A1D

IEL 6112451 — List of Inspection Equipment Numbers for Telescope, Elbow: M62

IEL 8243467 — List of Inspection Equipment Numbers for Telescope, Elbow: M62A1C

IEL 8572120 — List of Inspection Equipment Numbers for Telescope, Elbow: M62A1D

LSQAP 8243467 — Supplementary Quality Assurance List for Telescope, Elbow: M62A1C

LSQAP 8572120 — Supplementary Quality Assurance List for Telescope, Elbow: M62A1D

PACKAGING DATA SHEET

MIL–P–14232/P8243467 — Packaging of Telescopes, Elbow: M62 Series

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication. The telescope shall be manufactured in accordance with the drawings applicable to the model telescope specified in the contract or order and drawings pertaining thereto.

3.3 General specifications. The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL–O–13830 and MIL–F–13926.

3.3.1 *MIL–O–13820*

(a) Cleanliness.

(b) Illuminated reticle.

3.3.2 *MIL–F–13926*

(a) Order of precedence.

(b) Dimensions and tolerances.

(c) Inorganic protective surface finishes.

(d) Part identification marking.

(e) Workmanship.

3.4 Environmental.

3.4.1 *Storage temperature.* The telescope shall show no evidence of physical failure and shall meet all performance requirements of this specification at a standard ambient temperature of 60° to 90°F. after having been exposed to and thermally stabilized at ambient temperatures of —80° and +160°F.

3.4.2 *Vibration.* The telescope shall be vibrated at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (⅛ inch maximum excursion) for a period of 5 minutes ±15 seconds. Subsequent to vibration there shall be no loose or damaged parts and the telescope shall meet the requirements of 3.4.3 to 3.11 inclusive.

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3.4.3 *Cleanliness.* The telescope shall meet the cleanliness requirements of Specification MIL–O–13830. The surfaces of the reticle shall show no more than two particles of dirt. The area of each particle shall not be in excess of an area equivalent to a size two dig. There shall be no other dirt within the telescope which shall adversely affect observation and image quality.

3.4.4 *Sealing.* The telescope shall show no evidence of leakage when subjected to an external pressure of 5 pounds per square inch gage (psig) of dry nitrogen for a period of at least 5 minutes. The dewpoint of the nitrogen shall be at least as low as −25°F.

3.5 Collimation. The optical axis shall coincide with the geometric axis of the telescope within 0.5 mil as established by the exterior mounting surfaces.

3.6 Reticle alinement.

3.6.1 *Orientation.* When each of the three lines on the telescope body are successively in coincidence with a line inscribed on a telescope holding medium and with this line on the holding medium maintained in a true horizontal plane passing through the axis of the cylindrical mounting surface of the telescope, the reticle alinement of the telescope shall meet the requirements of 3.6.2.

3.6.2 *Horizontal reticle line.* The horizontal line (across the entire field of view) shall be horizontal within an angle subtended by 1.5 mils at one end of the reticle line when the other end of the reticle line is in coincidence with a horizontal target line (for the M62E2 elbow telescope the horizontal reticle line will be an imaginary line connecting the top of the graduations for the 160 mil angular spread).

3.7 Resolution. The resolving power at the optical axis shall be 15.0 seconds of arc or less.

3.8 Parallax. There shall be no more than 0.1 mil parallax between the image and the reticle, at the center of the field when a suitable target is viewed from a distance between 100 and 140 meters.

3.9 Eyepiece focus. The eyepiece focus shall be fixed between minus 0.75 and minus 1.0 diopter.

3.10 Eyeshield. The rubber eyeshield shall show no signs of deterioration, cuts, or tears and shall fit snugly on the telescope body.

3.11 Illumination. With the reticle illuminated, the reticle markings shall appear clearly defined when observed in a darkened area.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *General provisions.* The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of Specification MIL–I–45208 and the component inspection requirements of Specifi-

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cation MIL–F–13926. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL–STD–109.

4.2 Inspection provisions.

4.2.1 *Inspection lot.* Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Presentation of Lots and Drawing of Samples" specified in Standard MIL–STD–105.

4.2.2 *Examination and tests.* Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL–F–13926 and List of Supplementary Quality Assurance Provisions, LSQAP8243467 (Elbow Telescope: M62A1C) and LSQAP8572120 (Elbow Telescope: M62A1D). Examination and tests related to section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL–STD–105, and the inspection level and sampling plans specified in tables I and II titled "Classification Defects." Examinations and tests for packaging, packing and marking shall be in accordance with Specification MIL–P–14232 and section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual nonconforming items.

TABLE I. *Classification of defects*

Use Inspection Level II of Table III–A with
Sampling Plan Table IV–A of Standard MIL–STD–105

	Requirements	*Test procedures*
Critical: ***None.***		
Major: ***AQL 1.0% defective.***		
101. Cleanliness unacceptable	3.4.3	4.5.4
102. Improper collimation	3.5	4.5.5
103. Improper orientation	3.6.1	4.5.7
104. Incorrect reticle alinement	3.6.2	4.5.6
105. Insufficient resolution	3.7	4.5.8
106. Excessive parallax	3.8	4.5.9
107. Insufficient eyepiece focus	3.9	4.5.10
108. Loose or damaged eyeshield	3.10	4.5.11
109. Insufficient illumination	3.11	4.5.12
Minor: ***None.***		

TABLE II. *Classification of defects*

Use Inspection Level L6, of Table III–B with
Sampling Plan Table IV–A of Standard MIL–STD–105

	Requirements	*Test procedure*
Critical: ***None.***		

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Major: AQL 2.5% defective.

101. Vibration	3.4.2	4.5.3

Minor: None.

4.2.3 *Disposition of nonconforming product.* Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Disposition on Nonconforming Product" specified in Standard MIL–STD–105.

4.3 **Control tests.** One telescope of each model being procured shall be selected at random from each 200 produced, or from each months production, whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein, prior to conducting the following examinations and tests:

Control test	*Requirement*	*Test procedures*
Material	3.1	Appropriate Specification
Fabrication	3.2	Visual, Per Drawings
General Specification	3.3	Visual, MIL–F–13926 MIL–O–13830

4.3.1 *Control test for environmental requirements.* One telescope from each model being procured shall be selected at random from each 50 produced or each months production, which ever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

Control test	*Requirement*	*Test procedures*
Storage Temperature	3.4.1	4.5.1
Sealing	3.4.4	4.5.2

4.3.1.1 *Control test failure.* Should the control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 **Inspection equipment.** Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Inspection Equipment" requirement of Specification MIL–I–45208. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 *Government furnished inspection equipment.* Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with "Inspection Equipment," and "Government Furnished Material" requirements of Specification MIL–I–45208.

4.4.2 *Contractor furnished inspection equipment.*

4.4.2.1 *Government design.* All inspection equipment required by drawings form-

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ing a part of the contract and not provided by the Government shall be supplied by the contractor in accordance with the list of inspection equipment numbers, specified in 2.1 for the model being procured. Alternate designs may be substituted only as provided by "Inspection Provisions" of Specification MIL–I–45208.

4.4.2.2 *Contractor design.* The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of Specification MIL–F–13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.5 Test methods and procedures.

4.5.1 *Storage temperature.* The telescope shall be exposed at temperatures indicated in 3.4.1 by means of a cold chest and oven capable of maintaining these temperatures until the telescope is thermally stabilized. The telescope shall then be thermally stabilized at standard ambient temperature and shall meet the requirements specified in 3.4.1.

4.5.2 *Sealing.* The telescope shall be pressure tested by means of a pressure testing fixture, depicted on Drawing F7680673 and adapter, depicted on Drawing F7681049 in accordance with Instruction Sheet, Drawing F7680673, listed on appropriate IEL (listed in 2.1) for the telescope model being procured and shall meet the requirements of 3.4.4.

4.5.3 *Vibration.* The telescope shall be mounted in its normal operating position (as shown in the end item drawing) by means of Adapter, F7681053 supported on Fixture F7560085 listed on appropriate IEL (listed in 2.1) for the telescope being procured. The number of cycles, amplitude and duration of vibration shall be in conformance with the requirements outlined in 3.4.2. Upon completion of the vibration cycle the telescope shall be subjected to the examinations outlined in 4.5.3.1.

4.5.3.1 *Post vibration.* Subsequent to vibration, the telescope shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.4.2 prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts prior to being subjected to the remaining performance test. An audible examination shall be made by holding the telescope close to the ear and shaking it by hand to detect loose internal optical and mechanical parts. Any one defect shall be cause for rejection unless separate defect per hundred type sampling plans have been established and approved by the Government as a part of the contractors inspection system. Telescopes which have passed the above examinations shall be subjected to the following tests to determine conformance with 3.4.3.

4.5.4 *Cleanliness.* The optical surfaces shall be inspected for cleanliness by means of the shadowing method outlined in Specification MIL–O–13820 to determine conformance with the requirements of 3.4.3. The dig size shall be determined with the aid of Quality Surface Standard, Drawing C7641866.

4.5.5 *Collimation.* This test shall be conducted with the telescope positioned in fixture F7681056 after said fixture has been aligned and adjusted in accordance with the setup instructions provided with the fixture. The actual test is performed by sighting through the telescope under test

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and observing where the optical axis of the telescope under test falls upon the optical axis of the collimator. The optical axis of the telescope shall be considered as the intersecting point of the reticle centerlines. This intersecting point must fall upon the corresponding point of the collimator reticle within the limits shown on the collimator reticle. When the telescope is rotated 90°, each side of the original mounted position, the intersecting point must not exceed the tolerance limits provided on the collimator reticle in order to conform to the requirements outlined in 3.5.

4.5.6 *Horizontal reticle line.* When the telescope is rotated to an upright position, sight through the telescope and observe coincidence between the longest horizontal or verticle reticle line (depending upon the model being procured) and the corresponding collimator reticle line. When one end of the reticle line is placed in coincidence, the opposite end of the lines shall coincide within the limits of the tolerance block provided on the collimator reticle, in order to conform with the requirement outlined in 3.6.2.

4.5.7 *Orientation.* With the telescope positioned as outlined in 4.5.6, observe coincidence between the telescopes orientation marks and the corresponding line scribed in the text fixture. Repeat this procedure to determine that the remaining two orientation marks are within the limits specified on the telescopes body drawing. The test is made by alining either the telescopes horizontal or vertical reticle lines, parallel to the collimator reticle lines specified in 4.5.6. All orientation lines must aline with the fixture lines within the limits outlined on the drawing for the telescope body.

4.5.8 *Resolution.* With the telescope positioned as specified in 4.5.5 the test for resolution shall be as specified in Specification MIL–O–13830. The results of the test shall conform to the requirement outlined in 3.7.

4.5.9 *Parallax.* Loosen the fixture locking mechanism and aline a reticle line or mark on or near the optical axis of the telescope under test to a well defined reticle line or mark on the collimator reticle. Set the collimator objective lens to the distance specified in 3.8. Sight through the telescope and observe if parallax is present. Parallax is considered as apparent movement between the coincided reticle points while the observers head is moved from side to side, and up or down. The extent of parallax shall not exceed the limits specified in 3.8.

4.5.1.0 *Eyepiece focus.* This test is performed by using a three power dioptometer previously adjusted to the observers eye, with the objective assembly adjusted to remove parallax, and with the diopter index set to zero diopters. The test is made by placing a white background between the collimator and the telescope under test, and alining the dioptometers' optical axis to the telescopes optical axis and adjust the dioptometer to obtain the sharpest image of the telescope reticle while observing through the dioptometer. During this adjustment do not disturb the eyepiece adjustment. When the sharpest image is observed, the dioptometers scale shall indicate the telescopes dioptric setting. This setting must conform to the requirement outlined in 3.9.

4.5.11 *Eyeshield.* The eyeshield shall be visually inspected for conformance to the requirement outlined in 3.10.

4.5.12 *Illumination.* This test shall be performed by applying a 3 volt source of light to the appropriate aperture of the telescope. Sight through the telescope and block out all light from the objective end of the telescope. Proper illumination is assured when the observer can clearly distinguish all reticle lines and marks.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. The telescope shall be cleaned, preserved,

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packed and marked in accordance with Specification MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P8243467. The level of protection shall be as specified in the procurement document.

6. NOTES.

6.1 Intended use. Elbow Telescopes of the M62 Series are primarily intended for sighting for direct and indirect fire in both azimuth and elevation when used as component parts with various sightunits.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Selection of applicable level of preservation, packaging packing.

(c) Applicable Packaging Data Sheet number (see 5.1).

(d) Applicable stock number.

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